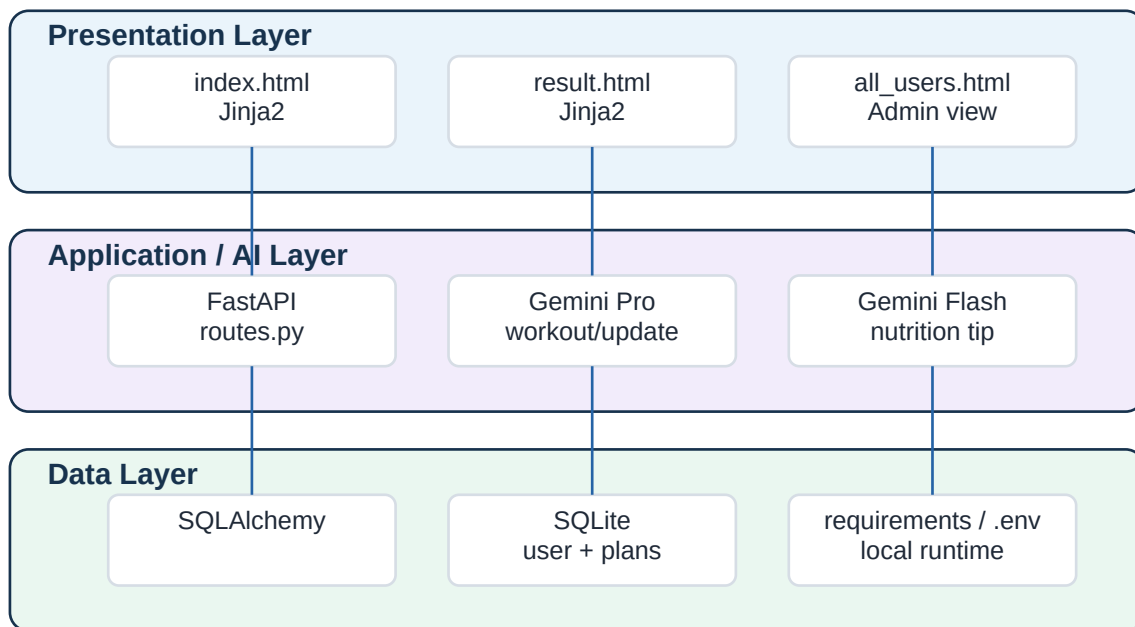


Solution Architecture

Date	01 October 2026
Team ID	6ab7f6f8c23107aaaa72871e
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Maximum Marks	5 Marks

Solution Architecture Diagram

The architecture follows the modular structure documented for FitBuddy: frontend templates, FastAPI routing, Gemini AI services and SQLite persistence.



User input -> FastAPI -> Gemini -> SQLite -> rendered result

Layer	Components	Responsibility
Presentation	HTML + CSS + Jinja2	Collects input and renders plans, tips, feedback and admin views.
Application / AI	FastAPI, Pydantic, Gemini Pro, Gemini Flash	Validates input, routes requests and generates / updates AI content.
Data	SQLAlchemy + SQLite	Persists user details and original / updated plans.